Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 6: Velocity–Time Graphs

One graph runs through Sections 1 to 3. The tram below pulls away from a stop, holds a steady speed, then brakes for the next one. Read the gradient for acceleration, read the area for distance, then use that distance for average speed — the same graph, asked three different ways. Section 4 is a new graph, and this time you draw it yourself.

You met the gradient formula on *Kinematics 5*, on a position–time graph, where it gave you a velocity. It is the same formula here. The only thing that changes is what is up the side of the graph — and that changes what the answer means.

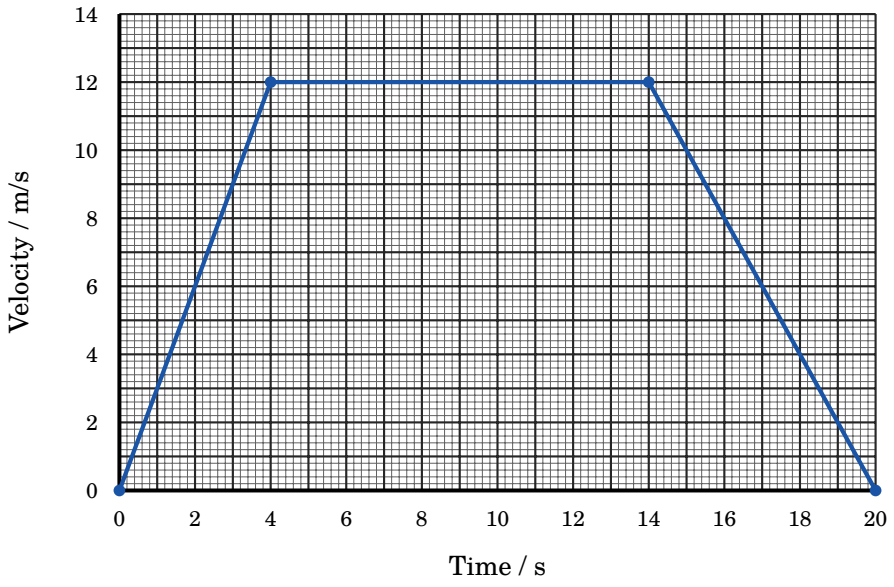
Section 1 — Gradient = acceleration

aP5-55

The gradient of a velocity–time graph is the acceleration. Same formula, one axis different:

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

Here the y values are **velocities** and the x values are **times**, so the gradient comes out as an **acceleration** in m/s^2. A flat line gives a gradient of zero: the velocity is not changing, so the acceleration is zero and the tram is still moving at a **constant velocity**.



Worked example — copy this shape

The first 4 seconds, as the tram pulls away. Read the two ends of that straight section off the graph:

point 1 = (0 s, 0 m/s) point 2 = (4 s, 12 m/s)

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{12 - 0}{4 - 0} = \frac{12}{4} = \mathbf{3\ m/s^2}$$

1. What is the tram's acceleration between **4 s and 14 s**? _____

2. What is the tram's acceleration between **14 s and 20 s**? Write down both points first. _____

Check your two answers against the shape of the line. The tram takes 4 s to reach full speed and 6 s to stop, so the braking slope is the *gentler* of the two — its answer should come out smaller in size than the 3 m/s^2 you started with, and it does. The flat section should give exactly zero.

Section 2 — Area = distance

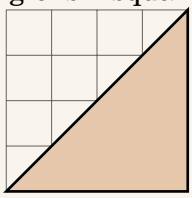
aP5-56, aP5-58, aP5-59

The area between the line and the time-axis is the distance travelled. Split a shape you do not recognise into ones you do — rectangles and triangles — work out each piece, then add them.

A second way to find the same area: count the squares. Count every square that is fully inside the shape. Then look at the squares the slanted edge cuts in half — each pair of half-squares, one on either side of the line, makes one more whole square. Add the two totals together.

Worked example — copy this shape

A small triangle, both ways. This triangle is 4 squares wide and 4 squares tall.



Formula: $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4 \times 4 = \mathbf{8}$

Counting squares: 6 whole squares sit fully under the diagonal, and the 4 half-squares it cuts pair up into 2 more whole ones. $6 + 2 = \mathbf{8}$.

Both methods agree. Use the formula when the shape is a clean rectangle or triangle; count squares when it is not, or as a check.

3. The tram, 0 s to 4 s. This is a triangle: 4 s wide, 12 m/s tall.

Distance $= \frac{1}{2} \times 4 \times 12 =$ _____

4. The tram, 4 s to 14 s. This is a rectangle: 10 s wide, 12 m/s tall.

Distance $= 10 \times 12 =$ _____

5. The tram, 14 s to 20 s. A triangle again — but a wider one than the first. Read its base off the graph before you start.

Distance = _____

6. Total distance travelled, the whole 20 s journey. Add your three answers above.

Total distance = _____

A quick scale check on the total. 180 m in 20 s at a top speed of 12 m/s is a sensible tram journey between two stops — not a sprint, not a crawl. If your total came out over a kilometre, or under ten metres, look for an arithmetic slip before moving on.

Section 3 — Average speed for non-uniform motion

aP5-43

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

Use the **total** distance — the one found from the area in Section 2 — divided by the **total** time on the graph. This is not the same as averaging the individual speeds you can read off the graph; it is the one number that describes the whole journey as if it had been travelled at a single, constant speed.

Worked example — copy this shape

The tram covers **180 m** in **20 s** in total (Section 2).

average speed $= 180 \div 20 =$ **9 m/s**

Check it against the graph. The tram's speed on this journey never drops below 0 m/s or rises above 12 m/s, so the average has to sit somewhere between those two numbers. **9 m/s does.** If your own average speed ever comes out higher than the fastest point on a graph, that is the signal to check the total distance or total time you used.

7. A different graph shows a cyclist covering **240 m** in **30 s**, made up of a speeding-up section, a steady middle, and a slowing-down section. What is their average speed for the whole 30 s?

_____

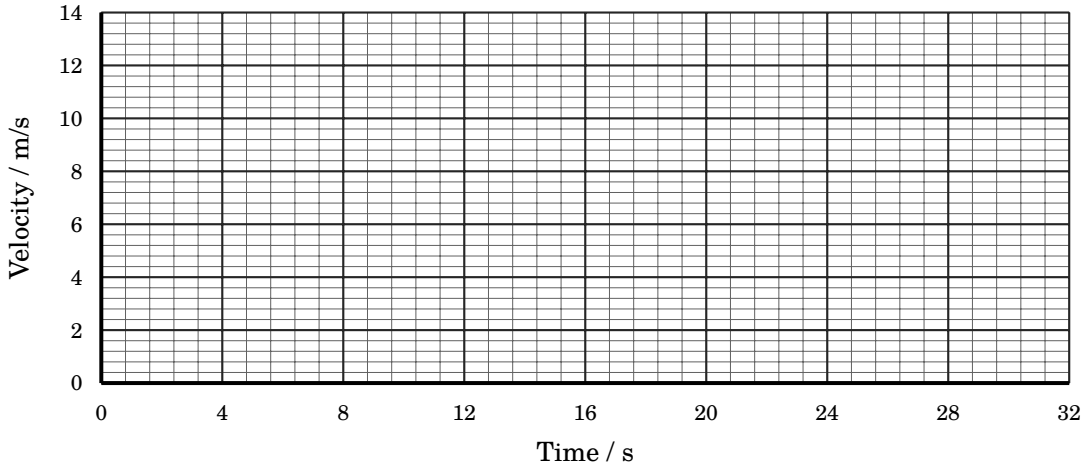
Section 4 — Your turn: the train

aP5-55, aP5-56, aP5-57, aP5-59, aP5-43

A train pulls out of a station, builds up speed in two stages, runs at line speed, then brakes into the next station. Its velocity was recorded every 4 seconds.

Time / s	0	4	8	12	16	20	24	28	32
Velocity / m/s	0	0	8	8	12	12	12	12	0

8. Plot the graph. Time on the horizontal axis, velocity on the vertical. Plot every point from the table, then join them up with a straight line between each pair — do not curve it. **Six straight sections** should appear.



9. What is the train's acceleration between **4 s and 8 s**? _____

10. What is the train's acceleration between **12 s and 16 s**? _____

11. What is the train's acceleration between **28 s and 32 s**? _____

12. Find the distance travelled in each section, using the area under your graph, and add them up. One section needs splitting into two shapes before you can do it.

Section	Shape under the line	Distance / m
0–4 s		
4–8 s		
8–12 s		
12–16 s		
16–28 s		
28–32 s		
Total		

13. How far did the train travel between the two stations? _____

14. Find the train's average speed for the whole 32 seconds. _____

Two checks on the total. The train's top speed was 12 m/s, so an average of 8 m/s sits sensibly below it — and the 16–28 s section alone is 144 m, more than half the journey, which fits: that is where the train is both quickest and running longest.